(a) Certificate-first task construction

(before model execution)

Feasible parent

fixed mission chains

Controlled intervention

capacities, SOC, windows, routes

Target & competitor

optimal target Z^*

feasible competitor Z'

Certification checks

- resource counterfactuals
- action restrictions
- independent replay

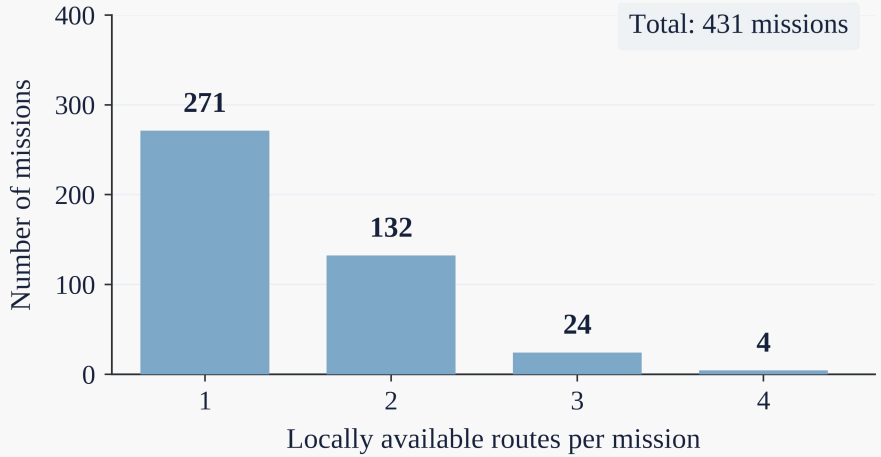
Admission & curation

- reject failed tasks
- structural quotas

Core-72

72 curated tasks

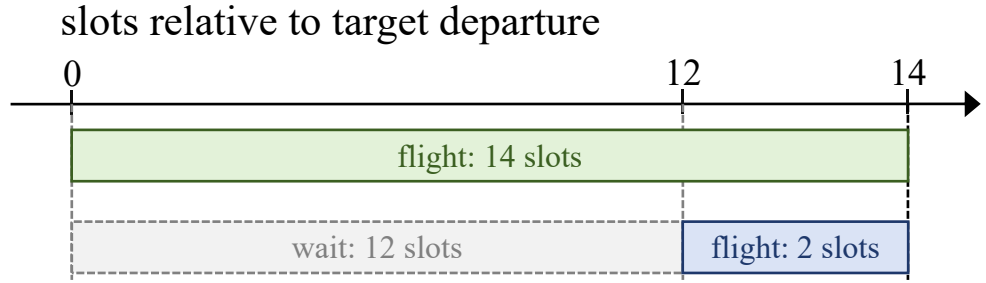
(b) Local route availability



before fleet-wide SOC and capacity checks

(c) Repeated L5 construction pattern

schematic · construction certificates



competitor - target, at equal L1 - L4:
+ 60 min release-to-departure time - 5 min charging = + 55 min L5